4 **(a)** Use the factor theorem to show that $(x-3)$ is a factor of $4x^3-8x^2-11x-3$. **[1]**

(b) Hence show that $4x^3-8x^2-11x-3=(x-3)(ax+b)^2$, where a and b are integers to be determined. **[2]**

(c) Sketch the graph of $y=4x^3-8x^2-11x-3$ on the axes in the Printed Answer Booklet. **[2]**

